

Introduction to the laboratory

This laboratory manual is designed to give an introductory knowledge on the practical aspects of Food Analysis. The aim is to deliver balanced analytical skills on how to determine the amount of major chemical components in foods and on their physico-chemical property changes relevant to food science and technology. A theoretical knowledge on the nature of water, acid base concept, solution concentrations, sugars, carbohydrates, amino acids, proteins, fatty acids, fats/oils, minerals, ash and fibers are important to understand the experiments. Students are expected to refresh their skills on the proper use of:

- Glass wares (burettes, pipettes, volumetric glass wares and beakers)
- Thermometers
- Balances
- Hot plates
- Water bathes
- Ovens
- Centrifuges
- Furnaces
- And so on

Introductory knowledge on titrimetric and gravimetric methods of chemical analysis is also important.

The following general safety instructions are important for the laboratory work. For details always follow the safety guidelines described under each experiment.

Before the laboratory

1. Carefully study the scientific principles upon which each experiment is based.
2. Write a flowchart for the procedures. The flowchart helps you to conduct the experiment efficiently.

In the laboratory

1. Follow instructions carefully.
2. Record your observations on your laboratory exercise book.
3. Do not over estimate yourself in the laboratory. In case you failed to understand the laboratory activities described, consult your instructor or the technical assistant.

After the laboratory

1. Interpret your experimental results.
2. Clean all the equipments and the table that you have used for the experiment.
3. Submit laboratory report before the next experiment is conducted.

Handling of reagents and solutions

1. Before using any reagent, read the labeling on the reagent bottle.
2. Select the best available grade of chemicals for analytical work. If there is a choice, pick the smallest bottle that will supply the desired quantity.
3. Replace the top of every container immediately after removal of reagent.
4. Reagent and solution bottle stoppers should not be left on desktops, where they may easily be contaminated with spilled chemicals.
5. Unless specifically directed, never return any excess reagents or solutions to the stock container.
6. Unless specifically instructed, do not insert spoons, spatulas, or knives in to a bottle containing a chemical reagent.

Glasses

Handle glass wares with care and if incase broken, dispose off the broken glass wares in special containers

Hot materials

Hot objects should not be placed directly on the desktop, but should be set on an asbestos pad or clean wire gauze. Handle moisture tins and crucibles used to dry and incinerate samples, respectively with tongs. Use test tube holders to handle test tubes. Handle hot beakers by use of towels. Tongs and forceps employed in handling heated objects should be kept scrupulously clean.

Flammable solvents

Do not let vapors to concentrate to flammable level in the work area, since it is nearly impossible to eliminate all chances of sparks from static electricity even though electric equipment is grounded. Use effective fume removal devices or fume hoods to remove these vapors when released.

Toxic vapor chemicals

Work under fume hoods and use masks to avoid inhalation of vapors.

Acids

Use effective acid-resistant fume removal devices wherever heating acids or performing reactions which liberate acid fumes. In diluting, always add acids to water. Keep acids off from skin and protect eyes from spattering. If acids are spilled on skin, wash immediately with large amounts of water.

Balances

In any chemical or food chemical analysis works, if mistakes are committed at the weighing stage, all the analysis ends up in wrong result. The following safety practices will assist you to minimize such errors.

1. Do not attempt to adjust the balance without the prior consent of the instructor or technical assistant.
2. Use analytical balance through the help of your instructor or technical assistant.
3. When the highest accuracy is required use analytical balance of highest degree of accuracy ($\pm 0.1\text{mg}$ or $\pm 0.01\text{mg}$).
4. When weighing, as far as possible, center the loads on the pans.
5. Use weighing boats if available, otherwise use clean small beaker. Some times solid chemicals can be weighed on leant free papers.
6. Mass measurement of volatile and hygroscopic materials requires special precautions. They should be weighed in a weight bottle.
7. Handle weights gently with special forceps only. Weights should never be touched with bare hand. Keep weights in their closed box.
8. Do not touch a dried object with bare hands.
9. Balance and case should be kept scrupulously clean. A camels' hair brush is useful for removal of spilled materials or dust.
10. The temperature of samples to be weighed must be at room temperature. If hot, keep in desiccators until the temperature is reduced to ambient temperature.

Volumetric apparatus

The most common apparatus used in the laboratory for volumetric determinations are the pipettes, burettes, measuring cylinder, volumetric flask, and conical flask of various sizes. For

reliable volume measurement, use of pipettes, burettes and volumetric flasks are advised. If the highest accuracy is required, uses of digital micro-pipettes or pipettes are advised. Pipettes and burettes have a graduation designed to deliver different specified volumes. However, volumetric flasks are designed to deliver specified one volume. Use conical flasks for titration purposes.

When a liquid is confined in a narrow tube such as burettes or pipettes, the surface exhibits a marked curvature, called meniscus. Volume reading must be taken at the bottom of the meniscus. When judging volumes, the eye must be in level with the bottom of the liquid meniscus, otherwise an error due to parallax will arise.

Only clean glass surfaces can support a uniform film of liquid. The presence of dirt or oil can cause breaks in this film. Thus, volumetric glass ware should be cleaned carefully before use.

In any scientific works, accurate measurements (mass, volume, temperature, etc...) plays a central role. After all an excellent scientist in any field is the one who makes an excellent accurate measurements.

Note

If student shall conduct unsafe laboratory practice, he/she will be reprimanded. If students are reprimanded twice, he/she can be suspended from further laboratory work and may receive an incomplete grade for the course.

Proper Methods for Cleaning Laboratory Glassware

Washing laboratory glassware is a bit more complicated than using simple soap and water.

Certain precautions must be taken in order to avoid destroying chemical solutions and experiments. Besides typical beakers, some other types of laboratory glassware include burets, pipets and volumetric flasks. These types of glassware all have special cleaning requirements.

Often, standard soap and water can ruin or spoil a chemical solution. Detergent specifically designed for washing laboratory glassware should be used to clean glass beakers. Additionally, microfiber, lint free cloths and cotton tipped applicator sticks should be used in place of standard dish cloths.

Detergents

When cleaning glass beakers and other laboratory glassware, it is generally recommended to use a proper detergent or solvent designed specifically for laboratory use. The following are just a few of the various detergents designed for the purpose of cleaning laboratory glassware:

- Alconox
- Liquinox
- Citranox
- Citrajet
- Solujet

These detergents are generally in a concentrated form and are often diluted to achieve the desired results. Lab glassware should be rinsed in the proper solvent, then rinsed a couple of times with distilled water, and finally, finished up with a couple of rinses with distilled water. Cotton tipped applicators in combination with microfiber cloths should be used to clean the inside of glass beakers.

Glass beakers and other laboratory glassware may require different cleaning methods depending upon the type of chemicals that were contained in the glassware. The following tips address methods of cleaning glass beakers that have contained various lab chemicals:

Water Soluble Solutions

Laboratory glassware that has contained water soluble solutions is perhaps the simplest to clean. Simply rinse the beakers or other containers 3 or 4 times in distilled water and allow them to dry.

Water Insoluble Solutions

For water insoluble solutions, the glassware should be rinsed 2 to 3 times with acetone or ethanol, followed by 3 to 4 rinses with distilled water. They should then be set aside and allowed to dry.

Acids and Bases: Beakers that have contained strong acids and bases should be cleaned under the fume hood. The laboratory glassware should be rinsed abundantly in tap water, followed by 3 to 4 efficient rinses with distilled water and set aside to dry.

Other tips

- * Glass beakers and laboratory glassware should be air dried to avoid scratching or damaging the surface. Towels or blowed air may also cause fibers or other contaminates.
- * It is generally easier to clean laboratory glassware directly following use, as it will be easier to remove all traces of chemical substances. Also, water and other liquids can ruin a chemical solution or experiment. It is therefore required that a dry glass beaker be used for each new experiment. By cleaning the glassware directly after use, it will allow for drying time.
- * If you must use glass beakers and other glassware immediately after cleaning, they should be rinsed 2 or 3 times with acetone. The acetone will evaporate quickly and remove any excess water.
- * When laboratory glassware has been cleaned properly, a distilled water rinse will form a smooth sheet when it is poured through clean glass beakers or other glassware. If you cannot see this sheeting action, you may need to apply more aggressive methods of cleaning.
- * Ground glass joints can be degreased by soaking a lint-free or microfiber cloth in either acetone or ether and wiping the joints carefully.

EXPERIMENT 1

Determination of moisture content in food samples

Objective

- ♦ To learn how moisture content in food samples are determined by air oven drying methods

Theory

Moisture content in food materials are measured for a number of reasons:

- (1) Because legal and labeling requirements urges to notify maximum or minimum amount of water in a given foods.
- (2) Microbial proliferation in foods depends on the available water in the foods.
- (3) Cost of shipping or transport is influenced by the amount of water in foods.
- (4) Texture, taste, appearance and stability of foods are influenced by the amount of water in the food.
- (5) Water influences the processing (mixing, drying, concentrating, flow properties or packaging) behavior of foods.
- (6) Any food chemical composition analysis demands the report should be based on dry matter basis.

Water in foods can exist in different states as: liquids, gases and solids. Because of heterogeneous nature of food materials, the physico-chemical environment of water in various foods is not of the same nature. Accordingly, one may find the following types of water in foods sample.

- (1) **Bulk water:** This is described as free water and each water molecule are surrounded by another water molecule. Physico-chemical property (melting point, boiling point, density, compressibility, heat of vaporization and electromagnetic absorption spectra) of this type water is the same as pure water.
- (2) **Trapped or capillary water:** Capillary water is held in narrow channels between certain food components by capillary forces. Trapped water is held within spaces of a food that are surrounded by a physical barrier that prevents the water

molecules from easily escaping, e.g., water in an emulsion droplet or in a biological cell. The majority of this type of water is involved in normal water-water hydrogen bonding and, thus they have similar physico-chemical properties to that of bulk water.

(3) **Physically bound water:** This type of water is in a molecular contact with other food chemical components (proteins, carbohydrates, lipids or minerals). The chemical bond between water molecules and these compounds are normally different from the normal water-water hydrogen bonds. Thus, this type of water has different physico-chemical properties from that of bulk water.

(4) **Chemically bound water:** Some of the water molecules present in a food may be chemically bound to other molecules as in the case of water of crystallization or as hydrates, e.g., $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$, $\text{C}_6\text{H}_{12}\text{O}_6 \cdot \text{H}_2\text{O}$. These bonds are much stronger than the normal water-water hydrogen bonds. Therefore, chemically bound water is in different physico-chemical environments as compared to that of bulk water hence its property also markedly different from bulk and capillary water.

As shown above, water molecules exist in a number of different molecular environments hence the physico-chemical properties of water are variable in different food samples. Thus, moisture content determination in foods is dependent on the conditions used and on the physico-chemical environment of water. Because of this, many methods of measuring moisture content are more sensitive to one type of water than to others. However, under a defined set of conditions the results are reproducible.

In this experiment, the moisture content in food sample will be determined as a loss in weight of a sample by heating under specified conditions. Samples like cereals will be heated at 130 °C for 1 hour in a forced air draught oven. If it is a dried vegetable or tea, heat in a vacuum oven for five hours at 100 °C using 100 mm of mercury. For meat and fish samples heating at 125°C for 2-4 hours is used.

Materials, Reagents or Chemicals,

1. Gravity-convection or mechanical-convection air oven maintained at appropriate temperature.
2. Moisture dishes or Petri dishes or moisture tins preferably 55 mm diameter and 15 mm height.
3. Desiccators containing absorbing material (anhydrous CaSO_4 , reignited CaO , alumina, molecular sieves type 4A, molecular sieves type 4A X W or P_2O_5). Silica gel and anhydrous CaCl_2 are not suitable desiccants.

4. Analytical balance
5. Tongs
6. Grain flour

Note: *use analytical balance through the help of technician! If there is a scarcity of materials the experiment can be conducted as demonstration.*

Procedure

Clean the moisture dish and dry at 130 °C for 1 hour and keep in a desiccator for about 15 to 20 minutes until it is cooled to ambient temperature (you may be given dried moisture dishes kept in a desiccators). Use tongs to handle the dish and do not touch directly with your bare hand. Take the mass of the dish on analytical balance and record the mass as " m_1 ". Weigh accurately about 2-3 g of representative (well mixed) flour sample in a triplicate (for student laboratory duplicate sample is enough) in to moisture

dishes and take the mass as " m_2 " (mass of sample + mass of moisture dish before drying). Sample should be spread evenly across the dish and weighed as rapidly as possible to minimize moisture loss. Dry the sample at 130 °C for 1 hour or at 100 °C for 6 hours. Care should be exercised not to overload the drying oven that may lead to an insufficient drying of an erroneous result. After drying is complete, remove samples from the oven and place in a desiccator to cool the sample to room temperature (for about 20 minutes) and weigh accurately and record the result as " m_3 ".

Calculation of moisture %

$$\text{Moisture}\% = \frac{m_{\text{initial}} - m_{\text{dried}}}{m_{\text{initial}}} \times 100$$

Where: $m_{\text{Initial}} = m_2 - m_1$ and

$$m_{\text{Dried}} = m_3 - m_1$$

Replicate determinations must check within 0.2% moisture; otherwise repeat the determination.

Questions

- (1) By air oven drying method which type of water can be removed from food samples?
- (2) Explain why milling of grain to fine particle size for moisture determination is required?
- (3) Describe why moisture determination in any food analysis is a primary task?

EXPERIMENT 2

Gravimetric method of total ash determination

Objective (s)

- ◆ To learn how total ash is determined by dry ashing of food samples
- ◆ To learn the skills of carbonization and ignition of samples

Theory

The inorganic residues in foods are determined as ash after the organic carbonaceous portion, water and other volatiles have been oxidized and evaporated away. The ash content shows the level of mineral residues such as Ca, K, Na and Cl and also indicates the food adulteration. If the ash content is larger than the normal range, it indicates addition of adulterants having a higher inorganic level than the food. If lower than the normal range, it indicates dilution of the food material with materials that have low inorganic level. Thus, determination of the ash and mineral content of foods are important to describe the nutritional labeling, nutritional quality (type and concentration of minerals), microbiological stability (high mineral content is some times used to retard the growth of microorganisms) and in processing, because the physico-chemical properties of foods are affected by the mineral content.

The ashing operation involves: getting of a representative sample, drying in an oven (100-113°C) for 1-2, carbonization over a Bunsen burner and ignition in a muffle furnace at 500-600°C until ashing is complete (12-24 hours). Some foods have peculiar ashing problems and demand some different treatments to get ashed. For example, foods of high moisture content demands drying to prevent spattering during ashing. Foods of high fat content demand the fat to be extracted with solvent or carefully smoked off before ignition on burner or at the lip of open muffle furnace. For sugar and sugar products addition of a few drops of pure olive oil is helpful to suppress the swelling. Other possible problems include contamination of samples by minerals in grinders, glassware or crucibles, use of deionized water is recommended when sample is prepared. On ashing, water and other volatile materials are vaporized and organic substances are burned in the presence of oxygen to CO₂, H₂O and N₂. Most minerals are converted to oxides, sulfates,

phosphates, chlorides or silicates. Some are volatile and may be partially lost, e.g., iron, lead and mercury. If analysis of such volatile minerals is desired, different ashing procedure such as wet ashing should be used.

Quartz Pyrex, porcelain, steel and platinum crucibles are used as a sample holder for ashing. However, selection of appropriate crucible depends on the type of sample analyzed and the furnace temperature used. Porcelain crucibles are widely used because they are inexpensive, can tolerate temperature up to $<1200^{\circ}\text{C}$ and are easy to clean. Porcelain crucibles are resistant to acids, but not recommended for alkaline samples because they can be corroded by alkaline samples. They also crack with rapid temperature changes.

If the sample appeared to not ash satisfactorily, breaking of the experimental sample into pieces with thick platinum wire is helpful. For samples such as flour-if failed to get ashed properly, addition of a few drops of water to the cooled dish, evaporation and re-ashed may be enough to obtain a white ash. If this appeared inadequate, cool the dish, add a few mL of water and filter through an ashless paper. Return the paper and contents to the ashing dish, dry and re-ash. As a last resort, the dish may be cooled and few drops of 10% nitric acid or 50% ammonium nitrate is added, evaporated and ashing is continued until white ash is obtained.

In this experiment, food samples are ashed after carbonization of the sample by the blue flame on a Bunsen burner.

Precautions

- (1) During carbonization take care to avoid sample loss due to swelling or spattering.
- (2) Do not touch the crucibles with your hand. Use clean tongs.
- (3) The furnace temperature is very high and while placing and removing crucible from the furnace use long clean tongs
- (4) Before removing the ashed sample from the furnace, reduce the furnace temperature gradually to $120-150^{\circ}\text{C}$
- (5) Immediately transfer the ashed sample with its crucible into the desiccator to cool the sample to room temperature before taking the mass on the analytical balance

Note: since ashing takes long time, after carbonizing the sample and placing it in muffle furnace students will leave with the technician to complete the ashing for them. The technician keeps

their sample in a muffle furnace. The next day the students will take the mass of their ashed sample.

Chemicals

Flour from grain and legume seeds

Materials

Analytical balance, desiccator, porcelain dish, Bunsen burner, drying oven, muffle furnace, tongs, iron ring, triangle or wire gauze, iron stand, clamp and bosshead.

Procedure

Clean the porcelain dish, dry at 120°C and ignite at 550°C in a muffle furnace for 3 hours. Remove the dish and cool in a desiccator. The technician is advised to conduct for the student before the laboratory. Take the mass of the crucible (ml) on analytical balance. Do not touch the crucible with your hand. Weigh 3 g of flour sample into a weighing porcelain dish and record the mass as m₂ (mass of sample plus mass of crucible). As deemed necessary dry the sample at 120°C for 1 hour in drying oven. Remove the sample dish from the oven and carbonize by blue flame of Bunsen burner (Figure 14.1) by placing the sample dish on wire gauze or triangle. Heat gently until the contents turns black.

Transfer the dish with its contents to a muffle furnace and ignite at about 550°C until free from carbon and the residue appears grayish-white (about 8 hours), Remove from the furnace and if appeared ashing is incomplete moisten with few drops of water. This helps to expose bits of unashed carbon. Re-dry in oven at 120°C for 1 hours and re-ash at 550°C until white ash color is obtained (few hours). Remove from the furnace and allow to cool for a moment. Place in desiccator until the temperature assumes ambient temperature and weigh (record the mass as m₃). Calculate and express % of ash on dry base as follows:

Calculation

$$Ash(\%) = \frac{m_3 - m_1}{m_2 - m_1} \times 100$$

Where: m₂-m₁ = sample mass in g on dry base and m₃-m₁ = mass of ash in g

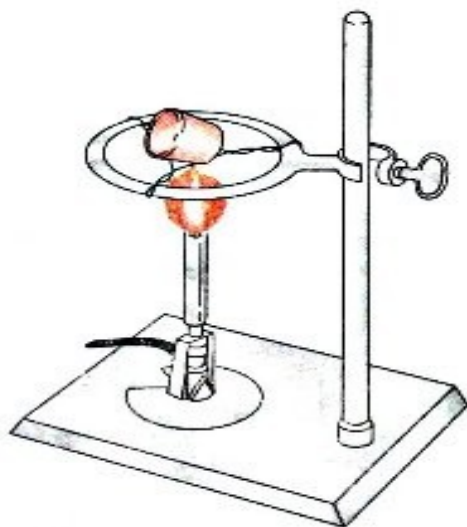


Figure14.1. Carbonization technique of sample on the blue flame

Questions

- (1) For the following food items: cereal grain, legume seeds, muscle and milk from which the mineral content that are determined as ash are contributed?
- (2) Why you use desiccator in this experiment?
- (3) Explain why the level of ash in food samples are analyzed?
- (4) What changes are taking place in the sample during carbonization and ignition?
- (5) What is spattering and swelling of the sample during carbonization?

EXPERIMENT 3

Acid base titration and determination of titratable acidity in foods

Objectives

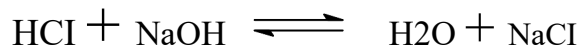
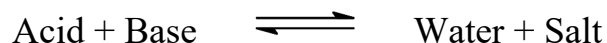
- ◆ To learn the skills of titrimetric or volumetric analysis by conducting acid-base titration
- ◆ To learn how the titratable acidity in food samples is determined

Theory

Titration is a quantitative method of analysis in which an equivalent quantity of standard solution (titrant) is added systematically from burette to a known volume of sample (analyte). The **equivalence point** of a titration is the point at which the reaction is theoretically complete. The point at which reaction is observed complete is called **end point** of the titration. The requirements of a titration are as follows:

- (1) The reaction must be stoichiometric. That is, there must be a well-defined and known reaction between analyte and titrant. In the titration of acetic acid in vinegar with sodium hydroxide, for example, a well-defined reaction takes place:
$$\text{HC}_2\text{H}_3\text{O}_2 + \text{NaOH} \rightarrow \text{NaC}_2\text{H}_3\text{O}_2 + \text{H}_2\text{O}$$
- (2) The reaction should be rapid. Most ionic reactions, as above, are very rapid.
- (3) There should be no side reactions, and the reaction should be specific. Titration demands the removal of an interfering substance. For example, in the above reactions, it demands no other acids present.
- (4) There should be a marked change in some property of the solution when the reaction is complete. This may be a change in the color of the solution or in some electrical or other physical property of the solution.
- (5) The end point should coincide with equivalence point or be at a reproducible interval from it.
- (6) The reactions should be quantitative. That is, the equilibrium of the reaction should be far to the right, so that a sufficiently sharp change will occur at the end point to obtain the desired accuracy.

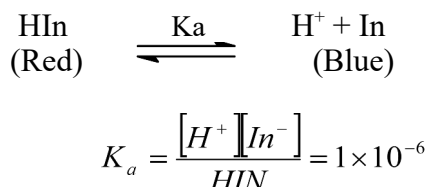
Acid-base titration involves a neutralization reaction in which an acid is reacted with an equivalent amount of base.



The titrant of acid-base titration is always a strong acid or a strong base. The end point of acid-base titration is monitored by using an acid base indicator (s) or by measurement of change in the pH of the solution with the pH meter (a potentiometer). A plot of pH change at different points of the titration gives the titration curve from which in the region of endpoint of the titration, large change in pH with small volume (0.02-0.03mL) of titrant added is observed. Figure 2.1 shows the titration of strong acid with strong base (A) and vice versa (B).

The indicator (s) used in the acid base titration are by nature organic weak base or weak acid usually consists of conjugated multiple bonds that absorb radiation in the uv-visible spectrum region. To serve as an indicator, color of the ionized form should be markedly different from the un-ionized form. One form may be colorless, but the other must be colored or both may be different colors. The pH range at which the visual color change of an indicator is observed can be accounted as follows. Consider the indicator (HIn) as weak acid in which the un-ionized form of the indicator is "red" and the ionized form is

"blue" colored. Take the dissociation constant of the indicator as $K_a = 1 \times 10^{-6}$.



By solving the equilibrium constant expression for [H+]

One can get

$$[\text{H}^+] = K_a \frac{[\text{HIn}]}{[\text{In}^-]} = 1 \times 10^{-6} \frac{[\text{HIn}]}{[\text{In}^-]}$$

$$\text{pH} = -\text{Log} [\text{H}^+]$$

$$pH = 6 + \log \frac{[In^-]}{[HIn]}$$

If a small quantity of this indicator was placed in a buffer solution, whose pH is 6,

$$6 = 6 + \log \frac{[In^-]}{[HIn]} \quad \text{or} \quad \log \frac{[In^-]}{[HIn]} = 0 \quad \text{or} \quad [In^-] = [HIn]$$

If pH of the solution is increased, $[In^-]$ will increase and the color of the solution will approach more nearly blue color of the $[In^-]$ ions. Our eyes can generally differentiate colors as different only if it is 10 times as intense as the other.

If the $\frac{[In^-]}{[HIn]} = 10$, which occurs at $pH = 7$, the color of the solution is not perceptibly different

from one whose pH is 10 i.e., when the ratio of $\frac{[In^-]}{[HIn]} = 10,000$. Likewise, if $\frac{[In^-]}{[HIn]} = 0.1$ ($pH =$

5), the red color of the solution would not be much different from $[HIn]$ solution alone at the same total concentration. The pH change is going from one color to the other, i.e., from pH 5 to pH 7 is 2. In general, most indicators require a transition range of about 2 pH units to show

a color difference. However, during the transition i.e., when $\frac{[In^-]}{[HIn]}$ is between 0.1 and 10

varying shades of red and blue Colors are obtained.

Table 2.1. Transition pH and acid to base color changes for some acid-base indicators

Common name	pH	Color*
Methyl orange	3.1 – 4.4	R - Y
Bromocresol green	3.8 - 5.4	Y - B
Methyl red	4.2 - 6.3	R - Y
Bromothymol blue	6.2 - 7.6	Y - B
Phenolphthalein	8.3 - 10.0	C-RV (faint pink)

* B = Blue; C = Colorless; R = Red; Y = Yellow; RV = Red violet

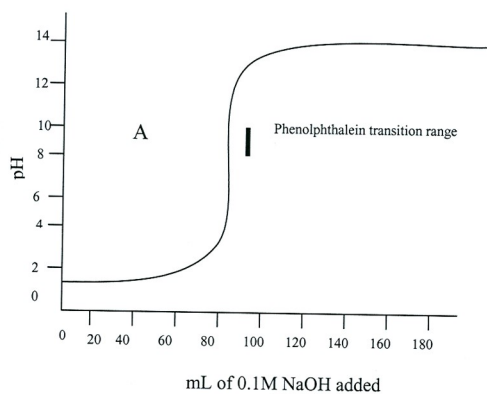


Figure 2.1. Titration curve of 100mL of 0.1M HCl with 0.1M NaOH

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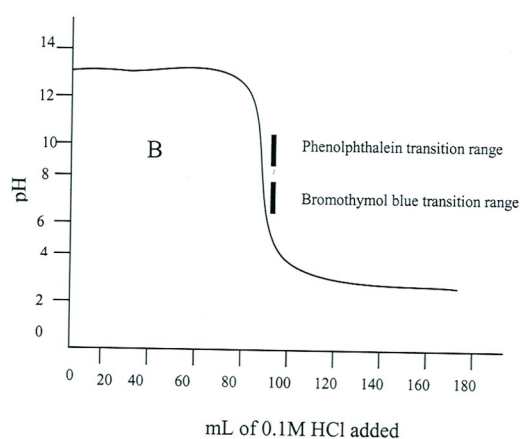


Figure 2.1. Titration curve of 100mL of 0.1M NaOH with 0.1M HCl

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In the titration for the determination of the amount of analyte, **standard solution**, whose concentration exactly known is used. There are two types of standard solutions:

- (1) **Primary standard.** This is prepared by dissolving an accurately weighed quantity of a highly pure material into accurately weighed volume.
- (2) **Secondary standard.** This is prepared by accurately weighing the compound to give targeted approximate concentration. Its more accurate concentration is then established by titrating it with the primary standard. For example, NaOH is not sufficiently pure to serve as primary standard. Thus, it is often standardized against a primary standard acid like potassium acid phthalate (KHP).

Chemicals that are used as a primary standard is required to meet the following requirements:

- (1) It should be 100.00% pure, although 0.01 to 0.02% impurities is tolerable if accurately known.
- (2) It should be stable to drying temperatures. And it should be stable indefinitely at room temperature. The primary standard is always dried before weighing.
- (3) It should be readily available
- (4) Although not necessary, it should have a high formula weight. That is a relatively large amount of it will have to be weighed to get enough to titrate. The relative error in weighing a greater amount of materials will be smaller than that for a very small amount.
- (5) If used in a titration, it should possess property features described for titrations mentioned above.

Titration technique

Standard solution is usually added from burette (often 50mL burette). The burette used must be scrupulously clean. Check that its valve is liquid-tight. Check the stopcock is rotating easily or not. Some times lubricating of the stopcock by grease is required (this is advised to be conducted by the technician). Avoid the grease not to clog into the stopcock and if so, remove the grease from inside the stopcock by using fine wire. If the stopcock is Teflon, it does not require lubrication. Check the tip for air bubbles and if any, work air bubbles to be out by rapid squirting of titrant through the tip or tapping the tip while solution is flowing. No bubbles should be in the barrel of the burette if burette is probably dirty.

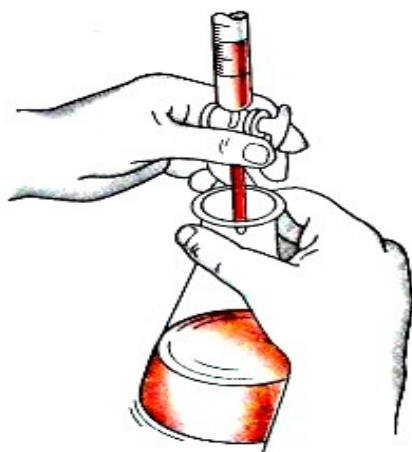


Figure 2.2. Proper technique of titration

The initial reading of the burette is taken by allowing it to drain slowly to the zero mark. Read the burette to the nearest 0.02mL (for a 50mL burette). The initial reading may be 0.00 ml or greater. Read at the bottom of the meniscus and avoid error due to parallax.

Measure the predetermined volume of the analyte into conical (Erlenmeyer) flask

(150-250ml), usually by using pipettes. Place the Erlenmeyer flask on a white background and positioned the burette tip into the neck of the flask. After adding of 2-3 drops of indicator add the titrant as an intermittent drop and swirl the flask with the right hand while the stopcock is manipulated with the left hand (Figure 2.2). This grip on the burette maintains a slight inward pressure on the stopcock to ensure that leakage will not occur. The solution can be also efficiently stirred by means of magnetic stirrer and a stirring bar (this arrangement is advised to be done by the technician).

As the titration proceeds, the indicator changes its color in the vicinity where the titrant is added because of local excesses. However, it rapidly reverts to the original color as the titrant is dispersed throughout the solution to react with analyte. When end point of the titration is approached, tendency to return to the original color occurs more slowly. When this starts to happen, stop the titration and wash the sides of the flask with distilled water from the wash bottle. A drop from the burette is about 0.05mL, and the volume is required to read to the nearest 0.02mL. It is therefore necessary to split drops near the end point. This can be done slowly turning the stopcock until a reaction of a drop emerges from the burette tip and then closing it. The fraction of drop is touched off onto the wall of the flask and is washed into the flask with distilled water from wash bottle. There will be sudden and permanent changes in the color that last at least 30 seconds at the end point of titration when a fraction of a drop is added.

Perform the titration in a triplicate. The subsequent titrations can be speeded up by using the first approximate volume reading of the end point of the first titration. After the titration is complete, unused titrant should never be returned to the original bottle, but should be discarded.

A. Acid base titration

Principle

In this experiment, HCl is standardized against primary standard borax. After establishing the concentration of HCl, it is used to titrate unknown concentration of NaOH. In the experiment, both acid base titration and standardization skills are learned.

Materials

Burettes (50mL), pipettes (10 and 25mL), Erlenmeyer flasks (150 to 250mL), beakers (250 to 500mL), funnels, reagent bottles (250 to 1000mL), indicator reagent bottles (50mL) and pasture pipettes, distilled water.

Chemicals

NaOH, HCl, borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) (equivalent weight 190.72), methyl red, bromothymol blue or phenolphthalein indicator, lemon, orange or mango juice.

Reagents (technicians are advised to prepare this and to standardize the unknown concentration of NaOH)

Borax (0.1N): Weigh accurately 1.9072g of borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$). Transfer to 100mL volumetric flask. Add about 30mL of distilled water and shake until dissolved. Adjust the volume to 100mL mark with distilled water.

Hydrochloric acid (approx. 0.1N): Dilute 5mL of concentrated (11-12N) hydrochloric acid to 500mL.

Sodium hydroxide: Weigh 4g of NaOH carefully. Dissolve it into previously boiled and cooled distilled water. Transfer it to 1L volumetric flask and dilute to the mark.

Methyl red: Dissolve 0.02g in 60mL of ethanol + 40mL of water.

Bromothymol blue: Dissolve 0.1g in 16mL of 0.1N NaOH + 234mL water.

Phenolphthalein indicator: Dissolve 0.05g in 50mL of ethanol + 50mL water. Procedure

For the technician

Fill in the burette with HCl and titrate it against a primary standard borax using methyl red indicator as follows. Pipette exactly 25mL of borax solution into 250mL Erlenmeyer flask. Add 2-3 drops of methyl red indicator and titrate with hydrochloric acid until the first red tint color appears marking end point of the titration. Repeat the titration three times. Calculate concentration of hydrochloric acid using the following expression:

$$N_{\text{HCl}} \times V_{\text{HCl}} = N_{\text{borax}} \times V_{\text{borax}} \Rightarrow N_1 V_1 = N_2 V_2$$

For example, if volume of HCl acid consumed to reach end-point of the titration is 23mL, the concentration of HCl is thus:

$$N_{\text{HCl}} = \frac{N_{\text{Borax}} \times V_{\text{Borax}}}{V_{\text{HCl}}} \Rightarrow \frac{0.1N \times 25\text{ml}}{23\text{ml}}$$

For the student

Fill in the burette with NaOH solution (unknown concentration). Pipette exactly 10mL of Standardized HCl (the technician will let you know the concentration of HCl) into 250mL Erlenmeyer flask. Add 2-3 drops of acid base indicator (methyl red, bromthymol blue or phenolphthalein). Titrate with NaOH to the endpoint marked by the color change of the indicator. Repeat the titration three times and calculate the normality of NaOH solution using typical expression given above.

B. Determination of titratable acidity in fruit juice Principle

To measure the amount of acid, it is necessary to titrate food sample with standardized sodium hydroxide solution. It is not sufficient to measure only pH of the food since this does not tell you the amount of acid present. In this experiment, the titratable acidity in fruit (orange, lemon or mango) juice extract is determined. The titratable acidity can be expressed as % acid/ml of liquid food or % acid/g of solid food.

Procedure

Take a known amount (*ca.* 1kg) of fruit, wash the surface and crush to extract juice. Record volume of juice extracted from 1kg of fruit. Make this data available for the student to use in the calculation. Dilute to 10% by taking 10ml of crude juice extract to 100mL with distilled water. Or if necessary 10g of fresh fruits can be mixed with hot water in a blender, cooked and filtered. All the above works are advised to be prepared by the technician in advance.

Pipette 10mL clear juice into 250mL Erlenmeyer flask. Add to it distilled water (5-10mL) and 2-4 drops of indicator. Titrate with NaOH (the concentration of NaOH should be established from the above titration and is probably *ca.* 0.1N) to the color change of the indicator. It is also necessary to know the major acid present, to establish the conversion factor for the calculation.

Conversion factors: citric acid = 0.064; tartaric acid = 0.075 and lactic acid = 0.09.

Calculation (calculate as % of citric acid)

A 1mL of 0.1N NaOH can neutralize 0.064g of citric acid.

$$\text{Citric acid}(\%) = \frac{[A \times 0.064 (\approx 0.1N \text{ NaOH}) \times B]}{C \times D} \times 100$$

Where A = average burette reading (mL of ~0.1N NaOH consumed), B = volume made with distilled water from which 10mL was taken, C = volume taken for titration (in this case 10mL) and D = mass of fruit from which the juice was extracted. In this it is assumed as if only 10mL

juice was obtained from 1 kg fruit. If you are not using the above formula, follow the following steps.

- (1) Calculate the mass of acid obtained in the 10mL volume used as analyte.
- (2) Using the mass obtained in (1) calculate what is the mass in 100mL from which 10mL was used for the titration of analyte.
- (3) Result (2) is obtained from 10mL of crude extract. Convert this mass to the total crude extract that is obtained from 1 kg fruit and then compute the %.

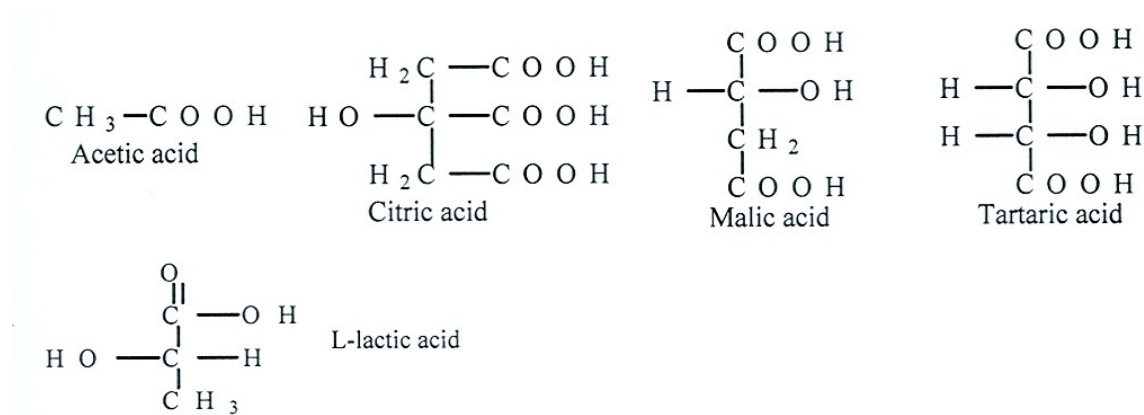


Figure 2.3. Structures of various food acids

Laboratory data collection format.

Replication	Standardization of HCl		Standardization of NaOH		Titratable acidity in fruit juice	
	Borax (mL)	HCl (mL)	HCl (mL)	NaOH (mL)	Juice volume (mL)	NaOH (mL)
1	25	?	10	?	10	?
2	25	?	10	?	10	?
3	25	?	10	?	10	?
Mean	25	?	10	?	10	?

Questions

- (1) The success of any laboratory analysis relay on the use of clean glass wares. Describe how you would you check glass wares like burettes, pipettes and Erlenmeyer flasks are clean or not? By referring any standard analytical chemistry or laboratory technique manuals, write the techniques used to clean glass wares?
- (2) Describe the different between end point of a titration and equivalent point of a titration?
- (3) What are the requirements for a chemical to be considered as a primary standard?
- (4) What is the molarity of a NaOH solution if 22.43mL of it exactly neutralized 25.00mL of a 0.1006 M HCl solution?

EXPERIMENT 4

Protein content analysis

(A) Kjeldahl method of crude protein analysis

Objective (s)

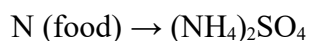
- ◆ To learn how crude protein content of food samples are determined by Kjeldahl method of nitrogen analysis
- ◆ To learn the process of digestion, distillation and titration in the course of Kjeldahl analysis
- ◆ To learn the colorimetric method of non-destructive ways of protein content analysis.

Theory

Proteins are polymers of nearly 20 amino acids. Proteins have many fold functionalities in foods: as source of energy and structural components as well as for other nonnutritional functionalities (gelling agents, emulsifiers, foaming agents, thickeners and stabilizers). Because of these and a number of other practical reasons, the crude protein content in a food samples are determined. Often the Kjeldahl method is used as standard and universal method for crude protein analysis in food samples. The method consists of three parts: digestion, distillation and titration.

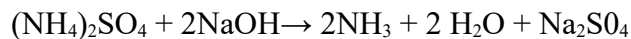
Digestion.

The food sample to be analyzed is weighed (*ca.* 0.1-1g) on an analytical balance into the digestion flask (round-bottom flask with a long neck, similar in appearance to a volumetric flask except for the round bottom and the lack of a calibration line). Then the sample is digested by addition of small volume (*ca.* 3-5mL) of concentrated H_2SO_4 (an oxidizing agent which digests the food), anhydrous Na_2SO_4 or K_2SO_4 that speed up the reaction by raising the boiling point of H_2SO_4 and a catalyst (CuSO_4 , selenium, titanium or mercury) to speed the reaction. Often 1g of catalyst mixture made of Na_2SO_4 or K_2SO_4 with anhydrous CuSO_4 in the ratio of 10:1 is used. Digestion converts any nitrogen in the food (other than that which is in the form of nitrates or nitrites) into ammonia and other organic matter to CO_2 and H_2O . In the acid solution, ammonia is not liberated as gas because it exists as an ammonium sulfate salt.



Distillation

After digestion is completed, the content in the flask is diluted by water and a concentrated NaOH (*ca.* 40%) is added to neutralize the acid and to make the solution slightly alkaline.



The ammonia is then distilled into receiving flask that consist a standardized strong acid (such as 0.1N H_2SO_4) or solution of excess boric acid (4%) for reaction with ammonia.

Titration.

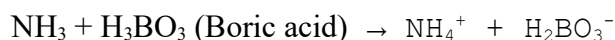
If H_2SO_4 is used in the receiving flask, the method is called back titration in which the excess acid is back titrated with standard NaOH.



Back titration reaction



But if boric acid is used, the titration is called an **indirect** titration, because the ammonia that is chemically equivalent in converting the boric acid (ammonia bound to an equivalent of borate ion) is directly titrated with standard acid (0.1N HCl).



In this experiment, preferably the borate method is used.



$$\text{Nitrogen}\% = \frac{V_{\text{HCl}} \text{ in } L \times N_{\text{HCL}} \times (\text{ca.} 0.1) \times 14.00 \times 100}{\text{samplweightondrybasis}}$$

Where V is volume of HCl in L consumed to the end point of the titration, N is the normality of HCl used often is about 0.1N and 14.00 is the molecular weight of nitrogen. To % of nitrogen is converted to % of protein by using appropriate conversion factors as follows:

% Protein = $F \times \% N$. Where, F is the conversion factor in most cases is 6.25. Use this factor in this experiment.

Note. Handle this experiment as demonstration.

Precautions.

(1) The digestion must be carried out in a fume hood, because thick fumes of SO_3 are evolved until the sample gets dissolved while the sample is digested at high temperature (*ca.* 350°C).

(2) Concentrated H_2SO_4 is extremely dangerous acid and avoid any spill. Wear face shield and heavy rubber gloves to protect against splashes.

(3) The reaction of NaOH pellets with water is an exothermic reaction. While preparing large volume of 40% NaOH use stainless steel initially because the heat generated can break glass beakers. After the temperature is reduced to room temperature transfer it into glass or plastic wares. NaOH is extremely caustic and can cause severe burns. Protect skin and eyes when working with it. Add pellets to water.

Equipment

Kjeldahl digestion flasks, analytical balance, weighing boat, digestion system, distillation unit, Erlenmeyer flask (250mL), burette

Sample. Flour from grain, other than wheat, If sheat is used change the conversion factor to 5.75 rather than 6.25.

Chemicals (all should be of analytical reagent grade)

Concentrated H_2SO_4 (96-98%), K_2SO_4 , Na_2SO_4 , CuSO_4 or NaOH pellets, HCl, H_3BO_3 , bromocresol and methyl red indicator

Reagents and their preparations

(1) *Sodium hydroxide* solutions (40%). Dissolve 40g NaOH into 100mL. For large volume that is usually required change the mass of NaOH accordingly.

(2) Boric acid (40%) with bromocresol green/methyl red indicator. Dissolve 40g of H_3BO_3 into 600mL boiled (still very hot) distilled water. Mix and add more boiled water upto 970mL and cool to room temperatures. Add to it 10mL of bromocresol green solution (10mg dissolved into 10mL alcohol) and 7mL of methyl red solution (10mg dissolved into 10mL alcohol). Finally, dilute to 100mL with distilled water and mix carefully.

(3) *Hydrochloric acid* (ca. 0.1N). Take 8.6mL from concentrated HCl solution (36.5-38.0%) and dilute to 1000mL mark. The exact concentration of HCl should be established by standardizing against a standard base.

Procedure

Weigh 0.3g of ground-dried sample into a Kjeldahl digestion flask. Add to it 1g of catalyst mixture (K_2SO_4 or Na_2SO_4 mixed with anhydrous CuSO_4 in the ratio of 10:1). Add 5mL of concentrated H_2SO_4 . Place the digestion flask in the digester and bring the temperature to 350°C . Digest for about 2 hours. Remove the flask from the digester and allow it

to cool. If the digestion is complete, the digested sample is colorless or light blue. If it is light brown or yellow the digestion is incomplete and in such case increases the duration of the digestion. After it is cooled, add carefully 30mL of distilled water into the digestion flask followed with 25mL (40% NaOH). Immediately distil the contents inserting the digestion tube line into the receiver flask that contains 25mL of 4% boric acid solution. Collect about 150mL of the distillate. Titrate the distillate against a standard acid (0.1N HCl). At the end point of the titration color is changed from green to steel-blue when half of a drop of acid is added. Taking the volume of HCl consumed from the burette reading, calculate the %N and %protein using the equation given above.

Note:

- (1) With Kjeldahl nitrogen analysis a control sample (sample that its nitrogen content is exactly established) must be analyzed along with the sample.
- (2) A blank sample is usually analyzed at the same time as the material being analyzed to take into account any residual nitrogen, which may be in the reagents used to call out the analysis. In that case subtract the volume of HCl consumed for blank titration from the volume HCl consumed in the sample titration for calculation.

(B) Biuret method

This is one of the non-destructive methods used in protein analysis. The method is based on the fact that, the nucleophilic-NH groups in proteins can form a purple colored complex with cupric (Cu^{2+}) ion under alkaline conditions. The complex has absorbance maxima at 330 and 545 nm. The absorbance at 545 is used for quantitative analysis of proteins. The method can be used to estimate proteins in solution in the range of 0.1mg/mL-2mg/mL.

Materials

Egg, flour or milk, spectrophotometer, volumetric flasks (250 and 500 mL) beakers, pipets, test tubes (>10mL)

Reagents

- (1) Copper sulfate solution. Dissolve 0.375g of copper sulfate into 250 mL distilled water.
- (2) Sodium potassium tartrate. Dissolve 1.5g of sodium potassium tartrate into 250mL.

(3) Sodium hydroxide. Dissolve 7.5g of NaOH into 250mL

Copper *Biuret* reagent: mix solution (1), Solution (2) and solution (3) If there is a precipitate, filter through fritted glass.

Procedure

Prepare a suspension of food proteins (milk, egg or flour) by taking 5.0g into 100 mL of distilled water. Filter through a filter paper or fritted glass. Then prepare a series of 5mL dilute protein solutions by taking 5mL (stock protein solution), 4mL (+ 1mL of distilled water), 3mL (+2mL of distilled water), 2mL (+3mL of distilled water), 1mL (+4mL of distilled water) and 0.0mL (+5mL distilled water). Take 0.5mL of protein solution and add to it 4.5mL of copper Burette reagent. Mix and allow 30 minutes to stand (violet color). Measure absorbance at 545 nm against blank (distilled water). Measure also the absorbance of sample from the blank. Plot absorbance versus concentration curve. From the calibration curve, determine the concentration of proteins in mg/mL?

Note if the reaction appeared not quantitative, check the concentration of protein whether is in the desired range. If not adjust the grams of proteins dissolved in the preparation of stock solution.

Questions

- (1) During the course of digestion of food material for Kjeldahl nitrogen analysis to which is assumed all nitrogen in food is converted?
- (2) Why additions of concentrated NaOH to the digest solution in the Kjeldahl nitrogen analysis are added?
- (3) What is the importance of distillation in the Kjeldahl method of analysis?
- (4) Describe the difference between back and indirect titration in the Kjeldahl method of nitrogen analysis?
- (5) In the course of digestion for crude protein analysis by Kjeldahl method how do you judge whether the digestion is complete or not?
- (6) What is the fundamental difference of protein content analysis in food samples by kjeldahl and Burette methods?
- (7) Draw the Burette reactions?
- (8) If you want to analyze the native enzyme levels in foods, which method do you recommend and why?

EXPERIMENT 5

Gravimetric method of lipid determination

Objective (s)

- ♦ To learn how crude lipid in food samples is determined.
- ♦ To learn the skills of solvent extraction, drying and gravimetric method of lipid mass determination.

Theory

Lipid compounds are distinguished from other major biochemical (proteins, carbohydrates, minerals and water) compounds by their solubility in non-polar solvents and insolubility in polar solvents. Lipids comprise the following group of compounds: glycerides ('fats/oil'), phosphoglycerides, waxes, terpenes, steroids, prostaglandins and glycolipids. The physico-chemical environment of lipid components: lipoproteins (lipids attached to proteins), glycolipids (lipids attached to carbohydrates). Because of this, method of lipid compound extraction in different food materials is all not the same.

Effective extraction of unbound fat/oil can be achieved by use of petroleum ether, n-hexane or diethyl ether or mixture of the two. The mixture of chloroform, methanol and water are often used to extract bound lipids because the polar solvent (water and methanol) is effective in disrupting hydrogen bonds to release bound lipids. In Rose-Gottlieb method (suitable for samples that contains proteins and sugars such as milk and milk products), the sample is treated with ethanol-ammonia (alkali) mixtures, and the fatty materials are extracted by solvent mixture of diethyl ether and petroleum ether. For food samples that consist of glycolipids and phospholipids, hydrolysis with 3M HCl for about 1 hour is used to release bound lipids prior to extraction with non-polar solvents (diethyl ether, petroleum ether and n-hexane).

Prior to solvent extraction of lipids, it is often necessary to dry and reduce the particle size of the sample. Drying is important because many organic solvents cannot easily penetrate into foods containing water. Size reduction is often done by grinding and is required to make the sample homogeneous and increase the surface area that can be exposed to solvent extraction.

Batch and semi-continuous solvent extraction techniques are often used in routine lipid determinations. In batch solvent extraction, sample and solvent in a suitable flask (like separatory funnel or large size screw capped test tubes) are vigorously mixed for some duration. The aqueous and organic phase is separated either by gravity or centrifugation. The aqueous phase is decanted off, and the lipid is determined gravimetrically after evaporating organic solvent. In this, the extraction should be done a number of times to improve efficiency of the extraction.

The Semi-continuous extraction method is commonly used for efficient and rapid extraction of lipids from food materials. *Soxhlet* method is the most commonly used technique and is an example of semi-continuous method. In this method, sample is placed in a porous thimble. The thimble is placed in an extraction chamber, which is suspended below a condenser and above a flask containing solvent. The flask is heated and the solvent evaporates and moves up into the condenser where it is converted into a liquid that drops into extraction chamber containing sample. Eventually, the solvent builds up in the extraction chamber and completely surrounds the sample. The design of the sample chamber allows the solvent to overflow and drops back down into the boiling flask when the solvent surrounds the sample exceeds a certain level. As the solvent passes through the sample, it extracts lipid and carries them into the flask. The lipids then remain in the flask because of its low volatility. At the end of extraction (6-12 hours), the flask containing the solvent and lipid are removed, the solvent is evaporated and the mass of lipid remaining is determined gravimetrically.

The *Goldfish* method is similar to the Soxhlet method except that the extraction chamber is designed, so that the solvent just drops through the sample rather than building up around it. This reduces the amount of time required to carry out the extraction, but has the disadvantages of channeling of the solvent, i.e., the solvent may preferentially take certain routes through the sample and therefore the extraction is inefficient. This is not a problem in the Soxhlet method, because sample is always surrounded by solvent. In this experiment, the Soxhlet method is used to demonstrate the extraction of a lipid.

Crude fat by Soxhlet method (handle this as demonstration for the student)

Materials

Soxhlet extraction apparatus, extraction thimbles (whatman cellulose), filter paper (Whatman No. 42), desiccator, evaporating dish, water bath, analytical balance

Chemicals

Food samples (oil seed or groundnut or maize flour) and solvent (petroleum ether, bp 40-60°C or anhydrous diethyl ether).

Procedure

Weigh accurately 3-4g of sample into a thimble lined with a circle of filter paper. Place thimble and contents into a 50mL beaker and dry in an oven for 2 hours at 110°C.

Transfer thimble and contents to extraction apparatus. Rinse the beaker several times with the solvent. Extract the sample contained in the thimble with the solvent in a Soxhlet extraction apparatus for 6-8 hours *at a* condensation rate of *at least* 3-6 *drops* per second. At the completion of the extraction, transfer the fat extract from the extraction flask into a pre-weighed evaporating small beaker (150-250mL) with several rinsing with the solvent. Place the evaporating small beaker in a fume hood and evaporate off the solvent on a steam bath until no odor of the solvent is detectable. Dry the beaker and contents in an oven for 30 minutes at 100°C. Remove from the extraction flask into a pre-weighed evaporating small beaker in a fume hood and evaporate off the solvent on a steam bath until no odor of the solvent is detectable. Dry the beaker and contents in an oven for 30 minutes at 100°C. Remove from the oven, cool in a desiccator and weigh beaker plus contents and calculate % of crude fat as given below.

$$\text{Lipid}\% = \frac{M_f - M_i}{\text{Sample mass by dry basis}} \times 100$$

Where “M_f” is mass of flask and lipid extracted; “M_i” is mass of dried flask.

Questions

- (1) Under what condition use of chloroform, methanol and water mixture as a solvent for extraction of lipid is preferable?
- 2) Why hydrolysis of sample with HCl is sometimes used prior of the lipid extraction?
- 3) Why sample drying and size reduction to fine size, are important before the sample is subjected to solvent extraction?

(4) Why non-polar organic solvents of low temperature boiling point are used for the extraction of lipids? What is the problem with the use of high temperature boiling point non-polar organic solvent?

(5) During the sample-dryings prior to analysis, low temperature drying is preferred (less than 125°C). What changes to the sample can happen if the sample is allowed to dry say at above 125°C for 3 hours?

EXPERIMENT 6

Determination of dietary fiber in food samples

Objective (s)

- ◆ To learn how dietary fiber content in food samples are analyzed and estimated.
- ◆ To learn the nature of various compounds collectively classified as fiber and principles on how they are determined.

Theory

Dietary fiber is a name given to a group of compounds that are not readily digestible by human digestive enzymes (*α*-amylase, protease and amyloglucosidase). This comprises non-starch carbohydrates (cellulose, hemi-cellulose, *B*-glucans, pectins, hydrocolloids and other pentosans), resistant oligosaccharides, resistant starches and lignin. Of these, lignin is not a carbohydrate and its detailed structure is not known, but believed is a condensation product of aldehydes such as vanillin, syringaldehyde and p-hydroxybenzaldehyde. Plant origin foods are valued for their high level of dietary fiber because cell wall is rich in dietary fiber.

Dietary fiber levels in foods are analyzed for a number of practical reasons. Dietary fiber is often classified as soluble and insoluble. Hydrocolloids (gums and mucilages) and pectic substances are classified as soluble dietary fibers whereas cellulose, hemicellulose and lignin are called insoluble dietary fibers. Both soluble and insoluble dietary fibers can be found in the same food even though their amount varies. Legumes, oats, citrus fruits are regarded as good sources of soluble dietary fibers, whereas wheat, rice, and many vegetables are rich in insoluble dietary fibers. To a limited extent soluble dietary fibers are digested to furnish very low calories to the body, while insoluble dietary fibers are excreted undigested and provides stool bulk, but not energy. Soluble dietary fibers are valued in suppressing the serum cholesterol level; whereas insoluble ones are beneficial because they speed the transit time and facilitate excretion of waste from the body resulting in suppressing of bowel tumor cells (bowel disorders from constipation and diverticulosis to cancer). In the current nutrition trend, high level of dietary fiber for healthy food market is encouraged. Thus, analysis of dietary fiber level in foods is often a requirement.

Because of diversity in the structure and properties of compounds described as fiber, no single analytical method is exhaustively comprehensive enough to meet all analysis demands that might be called for nutritional or other food technological application.

Currently available methods used in dietary fiber analysis can be categorized into three:

- (1) Nonenzymatic-gravimetric
- (2) Enzymatic-gravimetric (Southgate method)
- (3) Enzymatic-chemical methods (Englyst and Cummings method)

In the nonenzymatic-gravimetric methods (crude fiber, acid and neutral detergent fiber methods), defatted plant materials are extracted by boiling with dilute acid and followed by dilute alkali. The crude fiber (CF) method measures fiber as sum of lignin and cellulose. The acid detergent fiber (ADF) method measures fiber as sum of lignin, cellulose and acid insoluble hemicellulose. The neutral detergent fiber (NDF) method measures fiber as sum of lignin, cellulose and neutral insoluble hemicellulose. All these methods are not sensitive to estimate soluble dietary fiber level.

In the Southgate method, the defatted plant material is subjected to digestion with amylases, proteases and glucoamylases enzymes. The residue (sum of soluble and insoluble polysaccharides and lignin) that resisted digestion is measured as total dietary fiber (TDF). After TDF determination, further analysis can be made to classify as insoluble dietary fiber (IDF) and soluble dietary fiber (SDF).

In the Englyst and Cummings method, the defatted plant material is subjected to digestion with amylases, proteases and glucoamylases enzymes. The residue left is subjected to an acid digestion and the resulting monosaccharides are analyzed by chromatography (HPLC or GLC) or colorimetric methods.

In this experiment analysis method for CF by nonenzymatic-gravimetric is given. It is advisable to handle this experiment as demonstration!!

Method principle

In this, sample is digested with boiled H_2SO_4 (1.25%), vacuum filtered and washed. Subsequently digested again by boiling with dilute alkali (1.25% NaOH), vacuum filtered, washed and dried. The dried residue is ignited and CF is estimated as the loss in mass on ignition of the dried residue. In the system, fritted glass crucible is used for drying and ignition purposes. The method is applicable to grains, meals, flours, feeds and fiber-bearing material from which fat can be extracted

to leave workable residue.

Materials

(1) *Milling service* with 1-mm screen (No. 18)

(2) *Digestion system* with condenser to fit 600 mL beaker and hot plate adjustable to temperature that will bring 200mL water at 25°C to rolling boil in 15±2 minutes.

(3) Desiccator with efficient desiccant such as 4-8 mesh Drierite (CaCl₂ is not satisfactory)

(4) Filtering device with no. 200 type 304 or 316 stainless-steel screen easily washed free of digested residue. Either Oklahoma State filter screen or modified California plastic Buchner (consisting of 2-piece polypropylene plastic funnel), 70mm, without no.200 screen. Seal screen to filtering surface of funnel in advance if this arrangement is used, using small tip soldering iron.

(5) ***Filtration apparatus*** System to permit application of minimum vacuum necessary for filtration and washing of each sample within 3-5 minutes. Each unit consists of reservoir manifold connected to (1) H₂O aspirator through 120° stopcock, (2) atmosphere through second stopcock with metering device, and (3) receptacle cone shaped hard rubber gasket which provides vacuum seal with crucible. If necessary vacuum gage attached to manifold indicates vacuum applied to crucible. Crucible can be heated before and during filtration by flow of hot H₂O in surrounding jacket.

(6) ***Crucible***. Fritted glass, 50mL, *coarse* porosity. Clean as follows: brush and flow hot tap water into crucible to remove as much as possible. Submerge crucible in base cleaning solution (see below) ≥5 minutes, remove and rinse with hot tap water. Submerge in HCl (1: 1) solution (see below) ≥5 minutes, remove and rinse thoroughly with hot tap water followed by distilled water. After 3-4 uses, back wash by inverting crucible on hard rubber gasket in filtration apparatus and following near boiling water through crucible under partial vacuum.

(7) ***Liquid preheater*** for preheating water, 1.25% H₂SO₄ and 1.25% NaOH solutions to boiling point of water. Conveniently large heat resistant clean glass vessels arranged on heating mantle can be used or carefully heat on heating plate.

(8) ***Bumping chips or gl'ons***. Broken Alundum crucibles or equivalent granules can be used.

Reagents

- (1) Sulfuric acid solution. $0.255 \pm 0.005N$ (12.5g H_2SO_4 /1000mL distilled water).
- (2) Sodium hydroxide solution. $0.313 \pm 0.005N$ (12.5g NaOH/100mL distilled water free of Na_2CO_3 or nearly free of it)

For careful analytical data work, check the concentration by titration.

- (3) HCl (1+1) cleaning solution. Dissolve 250mL of HCl in 250mL of distilled water.
- (4) Base cleaning solution. Dissolve 5g Na_2H_2EDTA , 50g Na_2HPO_4 (technical grade) and 200g KOH in water and dilute to 1L. Storage in separate wide mouth containers holding 2-3L solution into which crucibles can be placed is convenient.

Determination

Sampling Get representative flour sample (about 2g). If possible use sample of flour with inherent low fat <1%. If the fat content is >1% de-fat the sample by extraction with ether or petroleum ether. Transfer sample into 600mL beaker avoid fiber contamination from paper or brush. Add 0.25-0.5g bumping granules, followed by 200mL near-boiling 1.25% sulfuric acid solution in small stream directly on sample to aid complete wetting of sample. Place beaker on digestion apparatus at 5 minutes intervals and boil exactly 30 min, rotating periodically to keep solids from adhering to side. Near end of refluxing place California bochner previously fitted with No. 9 rubber stopper to provide vacuum to about 25 mm Hg (735 mm pressure). At the end of refluxing flow near boiling water through funnel to warm it, then decant liquid through the funnel, washing solids into funnel with four 40-50mL portions near boiling water, filtering after each washing. Do not add wash to funnel under vacuum, but lift funnel from apparatus when adding wash.

Wash residues from funnel into reflux beaker with near boiling 1.25%NaOH solution. Place beaker on reflux apparatus at 5 minutes intervals and reflux 30 minutes. Near end of refluxing, turn on filtration apparatus, place coarse porosity crucibles in apparatus and adjust vacuum to about 25mm. Flow-near boiling water through crucible to warm it (keep near boiling water flowing through jacket during filtration and washing). At end of refluxing, decant liquid through crucible and wash solids into crucible with minimum of near boiling water. Increase vacuum as needed to maintain filtration rate. Wash residue once with 25-30mL near-boiling 1.25% sulfuric acid solution. Then, with two 25-30mL portions of near boiling water filter after each washing. Filtering and washing takes about 3-5 minutes/sample. Do not add wash to crucible under

vacuum.

Dry crucible with residue 2 hours at $130 \pm 2^\circ\text{C}$ or overnight at 110°C , cool in desiccator and weigh (w_1). Ash 2 hours at $550 \pm 10^\circ\text{C}$, cool in desiccator and weigh (w_2). Do not remove crucibles from furnace until temperature is $< 250^\circ\text{C}$, as fritted disk may be damaged if cooled too rapidly

Questions

- (1) Give at least three reasons why analysis of dietary fiber is essential in foods?
- (2) Classify dietary fibers as soluble and insoluble fibers?
- (3) Which group of fibers is determined as CF, ADF and NDF?
- (4) Write the structural features of cellulose, hemi-cellulose, gums and pectins?
- (5) What is resistant starch?
- (6) Why ignition of the dried residue is important and what is left as ignition residue?

EXPERIMENT 7

Physico-chemical tests on fats/oils

Objective (s)

- ◆ To learn some physico-chemicals tests that help to characterize fats/oils
- ◆ To learn the physical meanings of the tests and how each test are used in fats/oils processing industries to assure quality

Theory

Fats/oils (triacyl glycerols or glycerides) are esters of fatty acids and glycerol. Fats/oils are extracted either from plant or animal sources. Plant origin fats/oils are dominant products that are used in many food technological applications. As extracted from the source, crude fats/oils can comprise a diverse group of compounds (phospholipids, free fatty acids odor- and taste-imparting compounds, waxes, pigments, sulfur containing compounds, phenolic compounds, trace metal ions and trace agrochemical impurities), which are removed (i.e., their level reduced to an acceptable level) by refining operations. The properties of fats/oils are influenced by the nature and type of fatty acids esterified to glycerols, and by the level of impurities in the oil/fats. In this experiment some physico-chemical test that can help to identify the quality, identity and suitability of fats/oil for in some food end utilizations are given.

Materials

Plant seed (peanut, *nug*, sesame, any other available oil seed), refined oil, butter-fat, hydrogenated oils (hardened fats), thermometer, melting point determination apparatus, thermostat controlled water bath, refrigerator, funnel beakers (50, 100, 250, 500, 600 and 1000mL), filter papers, oil sample bottle or transparent bottle (50 to 150mL), glass rod.

(1) Melting point characterization

Natural Fats/oils do not possess sharp melting points because they comprise mixtures of triacyl glycerols. On the process of melting they pass through stages of softening before they get completely changed to liquids. The melting temperature is influenced by the degree of saturation in the fatty acids and the type of fat polymorph crystals. In this test the melting point temperature

of natural fat by capillary method under specified condition (i.e., the temperature at which the fats becomes perfectly clear and liquid) will be determined.

Procedure (handle as group work with brief demonstration)

In this, AccuMelt (automatic melt point measuring system) apparatus is used. Other apparatus arranged *by* use of thermometer, thermostat controlled water bath, rubber band and capillary tube can be also used. In the AccuMelt, the fiber optic transmitter projects a beam of infrared light onto the crystalline sample contained into the glass capillary tube. When a crystalline sample changes by melting, the amount of reflected light decreases. Then the computer measures this light intensity change as light intensity versus temperature.

Samples of fat to be used in the test are required to be clear and entirely free from moisture and impurities when melted. Because of this, samples are required to be taken from melted fats. Melt fat (5-10g) (butter, or hardened vegetable oil) sample in a beaker and filter through fluted filter paper. Dip capillary tubes in oil, so that fat in the tube stands about 1cm high. Fuse capillary tube and carefully by means of small blast flame avoiding air, and let cool. Check sample is liquid (if not re-melt over blue flame) and place the tube in a refrigerator (4-10°C) overnight. During the next day load the capillary tube sample into the AccuMelt by inserting into one of the sample position holes (1, 2 or 3) located on the top of the instrument. As the AccuMelt software "sees" the presence of the sample a sample window will be displayed asking for the sample number. Click on the text field and enter the sample number then close the sample window. After the sample has been loaded and the sample numbers entered, select the run button temperatures appropriate for the sample on the configuration menu. Select the run button to start the analysis. If the temperature of the block is below the initial temperature, the message "ramping up to initial temperature" will be displayed just to the right of the run button. If the sample is unstable with respect to the amount of light reflected off the surface the message "waiting for sample to stabilize" will be displayed. After the sample is stabilized and the initial temperature is reached or the computer determines that the sample is completely melted, the melting profile for the sample will be displayed along with the start melt point, melt point and finish melt point temperatures. This information is also displayed on the screen as a bitmap file.

(2) Cloud point

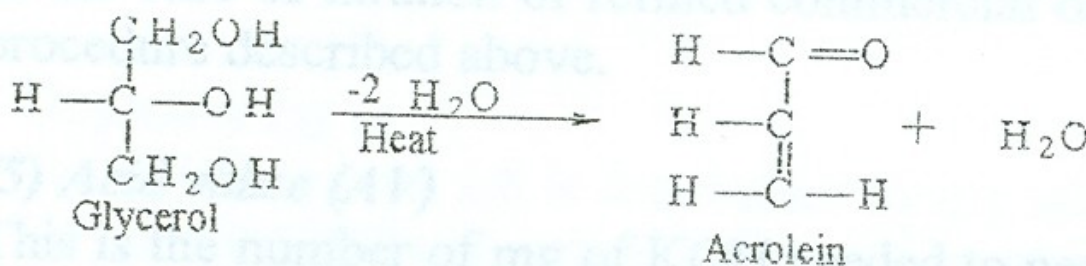
This test is used to estimate the crystallization temperature of animal and vegetable fats/oils, which is indicated by a visible cloud in the sample.

Procedure

Get one or two types of fats (animal or vegetable origin). Sample must be completely dry before test begun. Heat 60-70g fat in a baker to 130°C on a heating plate or heating mantle (use of heating mantle is more preferred) to completely destroy the fat crystal and cloud suspensions. Pour about 30 to 40mL of heated fat into oil sample bottle. Cool bottle and contents in a water bath (made of water and ice) contained in large glass beaker, stirring sample with glass rod (take care not to break the glass rod) enough to keep temperature uniform. Temperature used in the bath is advised to be not less than 2°C or no more than 5°C below cloud point. When sample is approached the solidification temperature, stir steadily and rapidly in circular motion to prevent super cooling and solidification of fat crystals on sides or bottom of bottle. From this point on, so not remove thermometer from sample, to avoid introduction of air bubbles that will interfere with test. Maintain test bottle in such position that upper levels of sample in bottle and water in bath are about the same. Remove bottle from bath and inspect regularly. Cloud point is the temperature at which the portion of thermometer immersed in oil is no longer visible when viewed horizontally through bottle and sample.

(3) Smoke point

Oil that are particularly used for food frying technological applications are required to have high smoke (200-230°C), flash and fire point temperatures. The smoke point of a fat or oil is the temperature at which its triacylycerols start to decompose in presence of air. Smoke is the sign of decomposition, mostly is due to the formation of acrolein from the decomposed glycerol of glyceride.



Acrolein is highly irritating and volatile aldehyde and is formed when glycerol is heated to the point where two molecules of water split from it. On prolonged frying the smoke point decreases and if it drops to less than 170°C the oil is no more suitable for food frying. In this experiment only the smoke point is tested because in most student laboratories flash and fire point determination is not advisable due to fire hazard.

Procedure

Arrange to heat sample in a well ventilated, well lighted place where the smoke can be seen against black paper. Add oil (10-20mL) (peanut oil, *nug* or any other commercial oil) to beaker (heat resistant glass) or dish until half full. Suspend a thermometer into the oil in a vertical position into the oil in a vertical position into center of the cup or beaker and heat the sample at the rate of 5-6°C/minutes on an electric heating plate. The formation of a stream of bluish smoke is an indication of the smoke point. Record the temperature. In some cases, slight puff appears before sample begins to smoke continuously. Ignore this.

(4.) *Determination of free fatty acids (FF A)*

The free fatty acids level in oils is often required to estimate the degree of rancidity in fats/oil and the level of alkali (NaOH) requirements in the refining operations. In this test crude extracted oil and commercial refined oil is titrated with standardized NaOH in the presence of neutralized ethyl alcohol.

Reagents

(1) *Ethyl alcohol* (95%). Denatured ethyl alcohol of the same strength can be also used. Neutralize the alcohol by adding enough 0.1N NaOH by adding 2mL phenolphthalein until faint permanent pink color appears.

(2) *NaOH* (0.1N and 0.25N). 0.1N NaOH-dissolve about 2g NaOH into 500mL distilled water. 0.25N-dissolve 5g of NaOH into 500mL distilled water. Titrate with standard acid

(3) *Phenolphthalein indicator*. Dissolve 0.05g in 50mL of ethanol + 50mL water.

(4) *Extraction of oils from oil seed*. Clean the seed. Crush and grind about 100g of the seed on the mortar and pestle and extract with n-hexane. Or boil the seed to 80-90°C with distilled water in a water bath and air dry. Extract the oil with screw press.

Procedure

Weigh 7-10 grams of fats/oils into 250mL Erlenmeyer flask. Add 50mL of neutralized alcohol and mix well and add 3 to 5 drops of phenolphthalein indicator. Titrate with standardized 0.25N NaOH with vigorous shaking until permanent faint pink color appears and persists for ≥ 1 minute.

$$\%FFA = \frac{mLNaOH \times NNaOH \times F \times 100}{Sampleweight \times 1000}$$

Where F is equivalent weight of FFA in which results are to be expressed. FFA is usually expressed as % oleic acid and equivalent weight is $282/1 = 282$.

In the case of titration of refined commercial oils, use 0.1N NaOH and follow the same procedure described above.

(5) Acid value (A V)

This is the number of mg of KOH needed to neutralize the organic acids present in 1g of fat/oil.

The free fatty acid (FFA) determination can be used in the calculation of acid value as follows:

$$Acidvalue = \frac{mLNaOH \times NNaOH \times 56.1}{sampleweight}$$

Where 56.1 is equivalent weight of KOH.

Questions

- (1) Describe what factors influence the melting point of fats?
- (2) Explain the difference between mixed and simple glycerides?
- (3) In the melting point determination why sample is required to be free of moisture?
- (4) What is fat crystal polymorph and which type of fat crystal polymorphism is desirable in fats?
- (5) How the nature of fatty acids nature can influence the cloud point?
- (6) Define the smoke, flash and fire point in fats/oils?
- (7) Why is the smoke point is desirable to be high? If the smoke point of frying oil drops what changes you anticipate take place in the oils?
- (8) What is rancidity and describe what type of rancidity reactions are expected in fats/oil?
- (9) If the acid value is large what does it indicate?